

18: Tips for Case Studies

Stat 230 - Spring 2026

```
library(tidyverse)
library(ggformula)
library(ggResidpanel)
library(bayesrules)
library(patchwork)
```

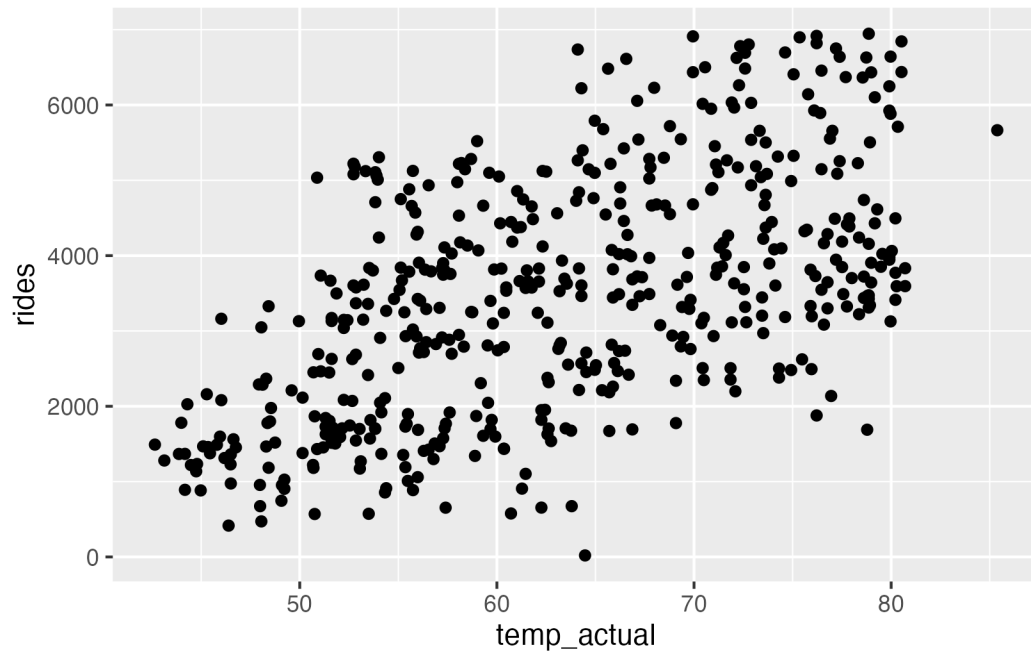
```
mod <- lm(rides ~ temp_actual, data = bikes)
```

1 Figure size

In the first few lines of your code chunks, you can specify figure output.

Default:

```
gf_point(rides ~ temp_actual, data = bikes)
```

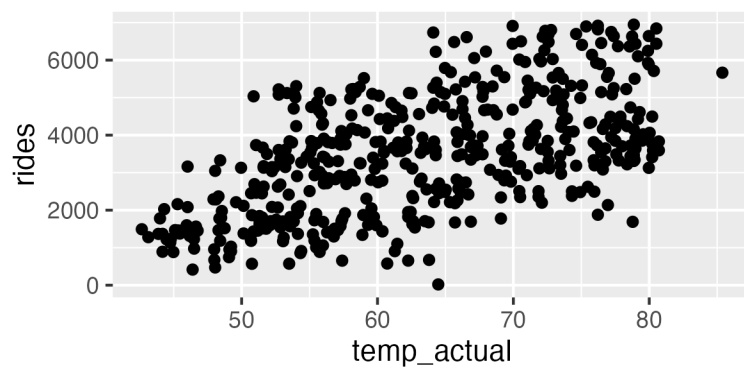


with

```
#l fig-width: 4  
#l fig-height: 2
```

at top of chunk:

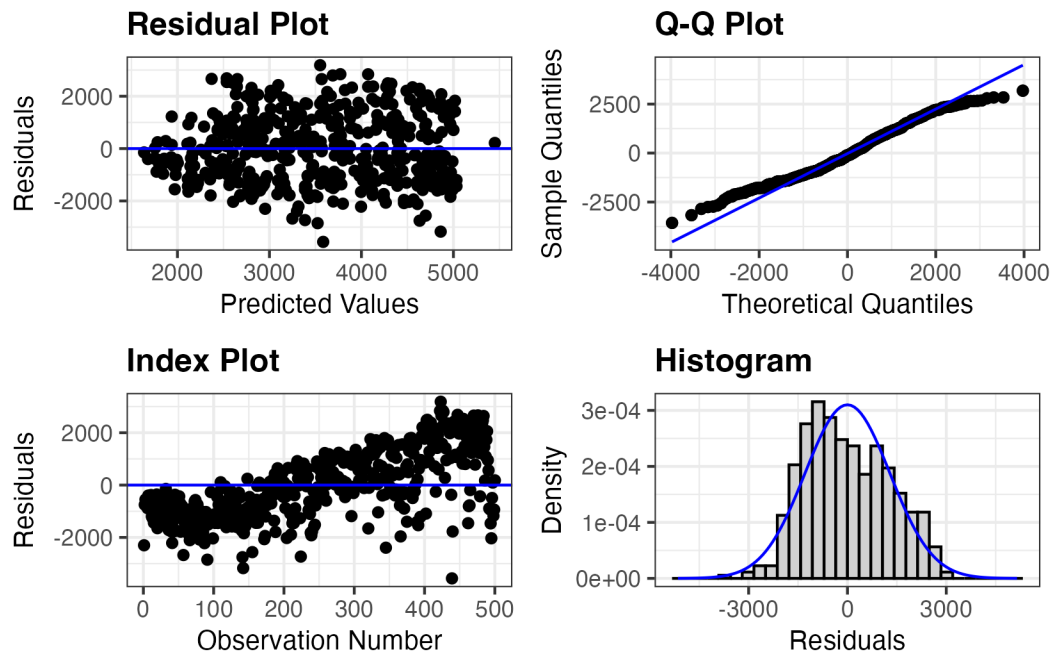
```
gf_point(rides ~ temp_actual, data = bikes)
```



2 Choosing a couple of residual plots

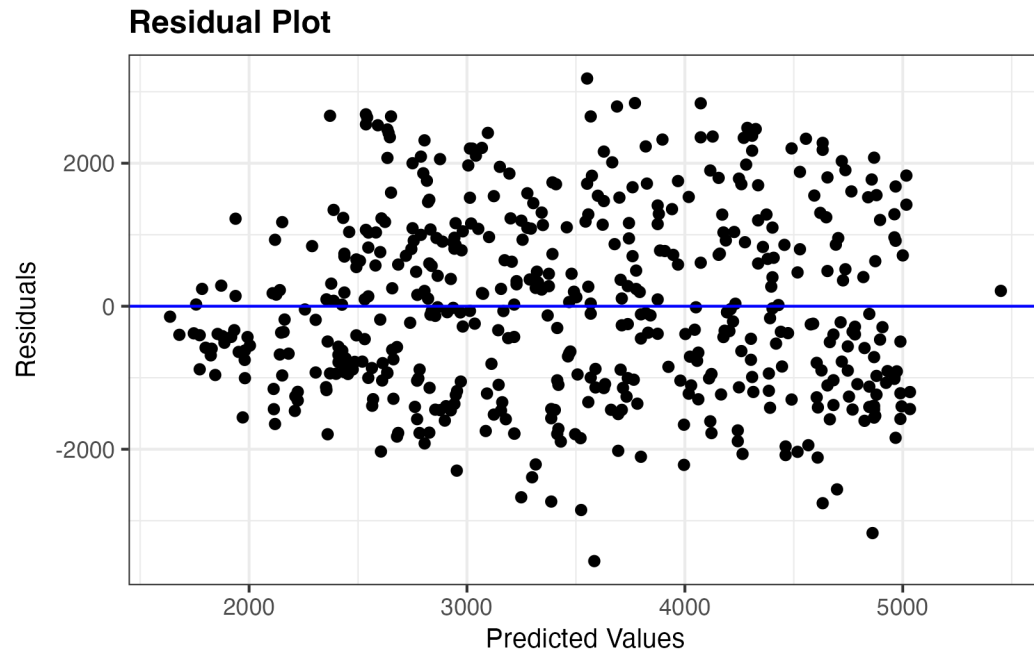
Sometimes, you might want to include some residual plots instead of the full panel (For example, you want to mention a weird trend in the index plot but all other residual plots are uninteresting). Specify using `plots` argument.

```
resid_panel(mod)
```



vs

```
resid_panel(mod, plots = "resid")
```

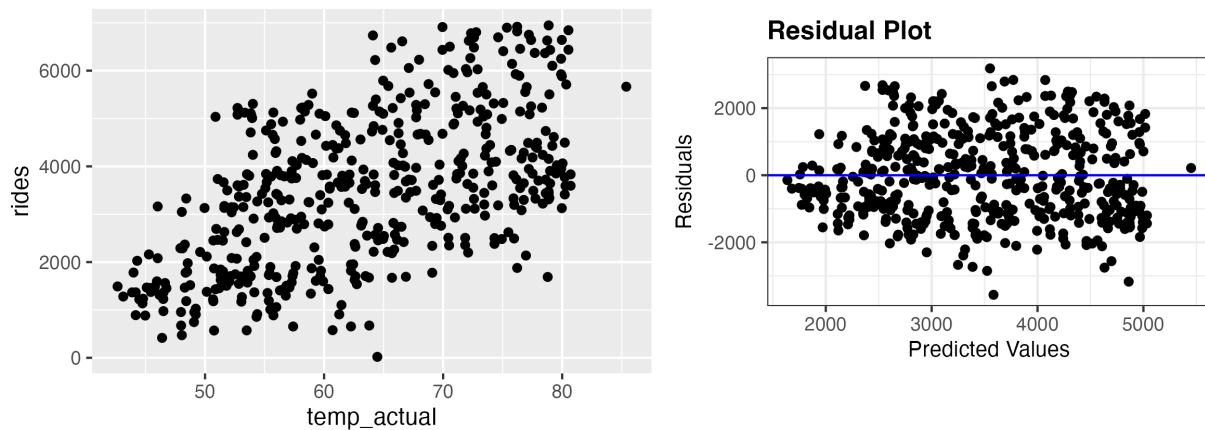


3 Combining graphics

Use the `{patchwork}` package to save individual plots and put them side by side

```
p1 <- gf_point(rides ~ temp_actual, data = bikes)
p2 <- resid_panel(mod, plots = "resid")

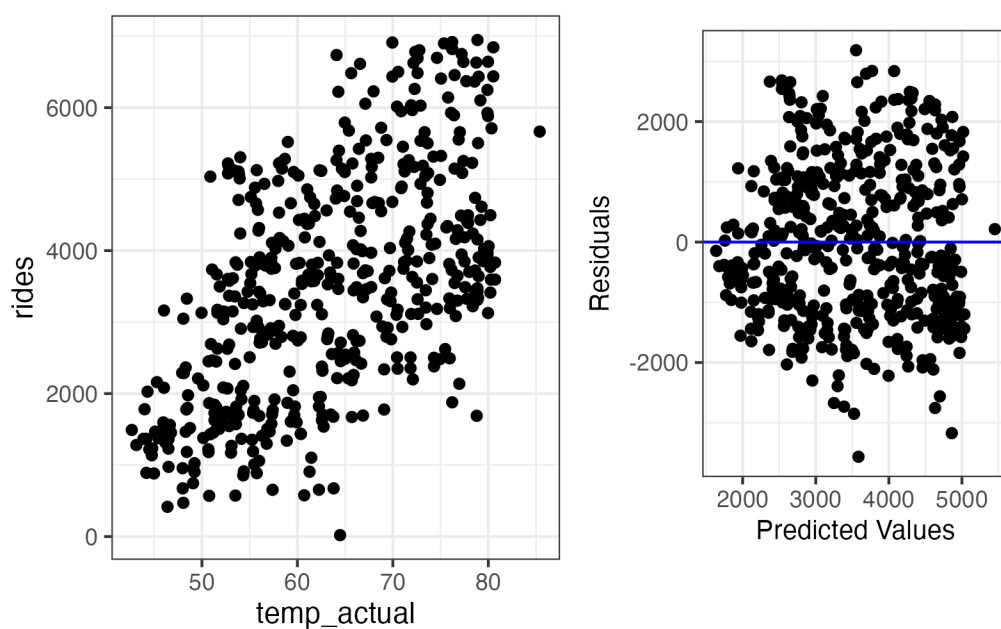
p1 + p2
```



4 Customize theme

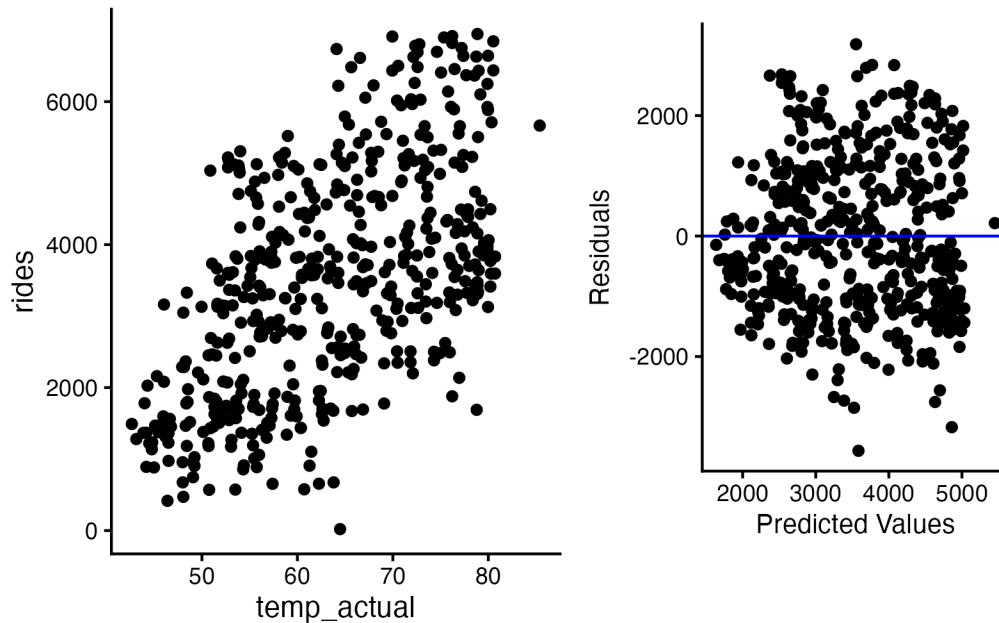
Oh no! The plots above look terrible together because they use different themes. *Note:* setting a theme is different with “regular plots” compared to resid panels

```
p1 <- gf_point(rides ~ temp_actual, data = bikes) + theme_bw()  
p2 <- resid_panel(mod, plots = "resid", theme = "bw", title.opt = FALSE)  
  
p1 + p2
```



Classic theme:

```
p1 <- gf_point(rides ~ temp_actual, data = bikes) + theme_classic()  
p2 <- resid_panel(mod, plots = "resid", theme = "classic", title.opt = FALSE)  
  
p1 + p2
```



Note: I know they show up as different sizes. I am not aware of a workaround besides using `augment()` and making your own residual plots.

5 Tables

The default summary output takes up too much space in a case study report:

```
summary(mod)
```

Call:

```
lm(formula = rides ~ temp_actual, data = bikes)
```

Residuals:

Min	1Q	Median	3Q	Max
-3564.3	-1009.3	-116.4	959.1	3184.6

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2168.77	363.64	-5.964	4.67e-09 ***
temp_actual	89.23	5.67	15.739	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1288 on 498 degrees of freedom

Multiple R-squared: 0.3322, Adjusted R-squared: 0.3308

F-statistic: 247.7 on 1 and 498 DF, p-value: < 2.2e-16

Instead, use `broom::tidy()` for a more compact output.

```
broom::tidy(mod)
```

```
# A tibble: 2 x 5
```

	term	estimate	std.error	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-2169.	364.	-5.96	4.67e- 9
2	temp_actual	89.2	5.67	15.7	1.35e-45

You can also specify confidence intervals:

```
broom::tidy(mod, conf.int = TRUE)
```

```
# A tibble: 2 x 7
```

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	-2169.	364.	-5.96	4.67e- 9	-2883.	-1454.
2	temp_actual	89.2	5.67	15.7	1.35e-45	78.1	100.

And to make it look EVEN nicer, pass it into `knitr::kable()`:

```
broom::tidy(mod, conf.int = TRUE) |>  
knitr::kable()
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2168.77362	363.641906	-5.964036	0	-2883.2350	-1454.3122
temp_actual	89.23389	5.669703	15.738723	0	78.0944	100.3734

which has a `digits` option:

```
broom::tidy(mod, conf.int = TRUE) |>
  knitr::kable(digits = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2168.77	363.64	-5.96	0	-2883.24	-1454.31
temp_actual	89.23	5.67	15.74	0	78.09	100.37

6 Equations

If you are doing your report in .qmd, you might want to include some math! To include an inline equation, use single $\$: y = \beta_0 + \beta_1 x_1 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$.

To give an equation its own line, use double $\$$

$$y = \beta_0 + \beta_1 x_1 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$$

(this typesetting language is called LaTeX and there are a bunch of resources available online)

- <https://latexeditor.lagrida.com/>
- <https://webdemo.myscript.com/views/math/index.html#>